

VAPT LAB ENVIRONMENT

Lab Setup Documentation

Kali Linux • DVWA • Metasploitable 2

Isolated Virtual Network: 192.168.100.0/24

Prepared as a practical lab setup record for authorized VAPT training

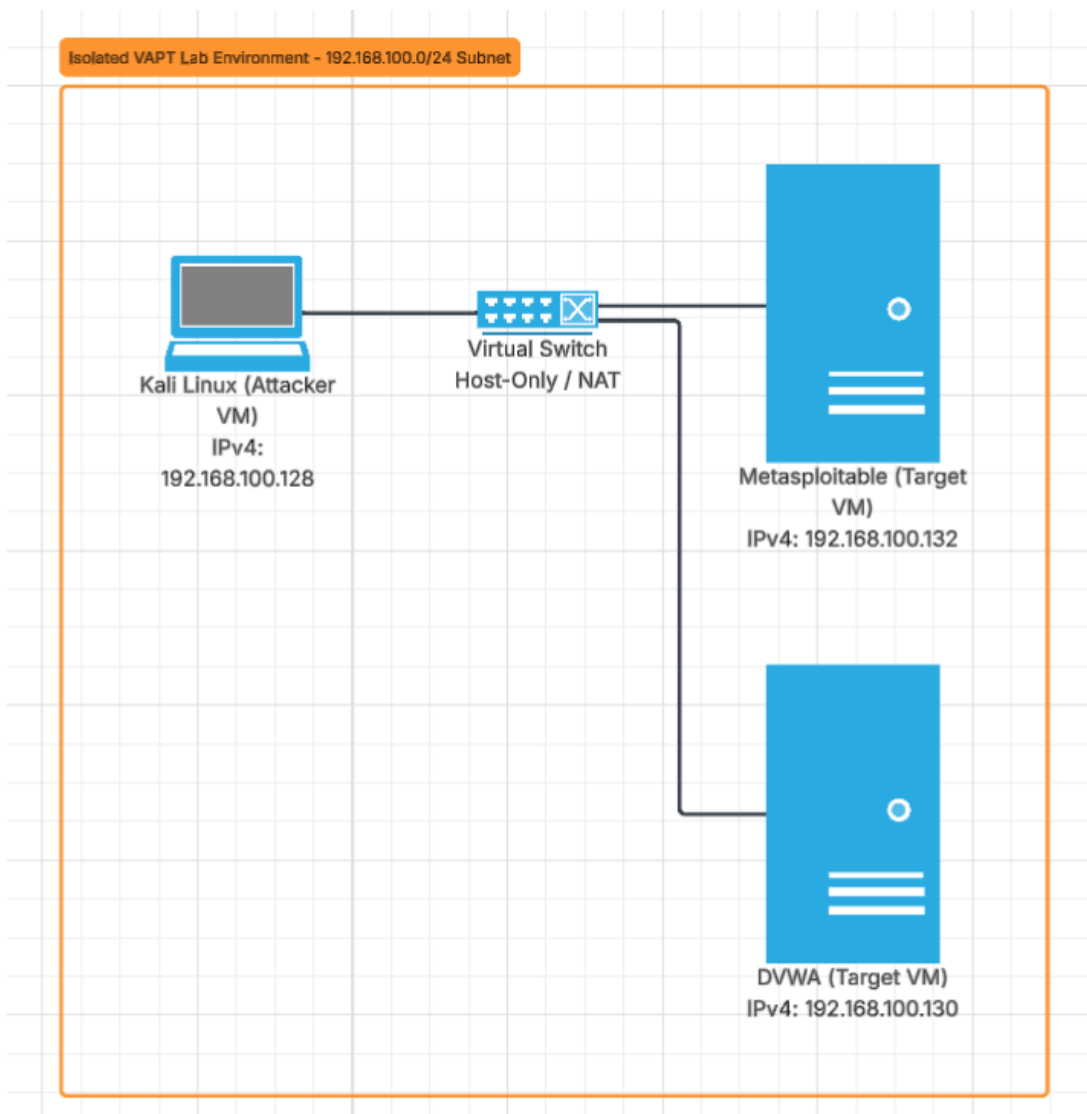


Figure 1. Isolated VAPT lab topology and addressing plan.

1. Lab Overview

This document records the setup of an isolated Virtual Penetration Testing (VAPT) lab built with VMware Workstation. The environment contains a Kali Linux attacker virtual machine and two intentionally vulnerable targets: Damn Vulnerable Web Application (DVWA) and Metasploitable 2.

The machines are connected to the same private 192.168.100.0/24 network so that security testing can be performed without exposing the intentionally vulnerable systems to the public Internet.

2. Lab Objectives

- Create an isolated virtual network for cybersecurity training.
- Connect the Kali Linux attacker VM to the target VMs.
- Configure and verify IP addressing on the Ubuntu/DVWA and Metasploitable systems.
- Confirm network reachability from Kali Linux to both targets.
- Verify that DVWA is accessible from the Kali browser.
- Maintain a controlled environment suitable for authorized vulnerability-assessment exercises.

3. Lab Components and Addressing

Virtual Machine	Role	IPv4 Address	Network
Kali Linux	Attacker / Testing VM	192.168.100.128	192.168.100.0/24
Ubuntu 26.04 / DVWA	Web application target	192.168.100.130	192.168.100.0/24
Metasploitable 2	Vulnerable target	192.168.100.132	192.168.100.0/24

The screenshots confirm 192.168.100.130 on the Ubuntu/DVWA host and 192.168.100.132 on Metasploitable 2. The topology diagram identifies Kali as 192.168.100.128.

4. Step 1 — Configure the Virtual Network

A private VMware network was used to connect the three virtual machines. The lab topology places Kali Linux, the Ubuntu/DVWA target, and Metasploitable 2 on the same 192.168.100.0/24 subnet.

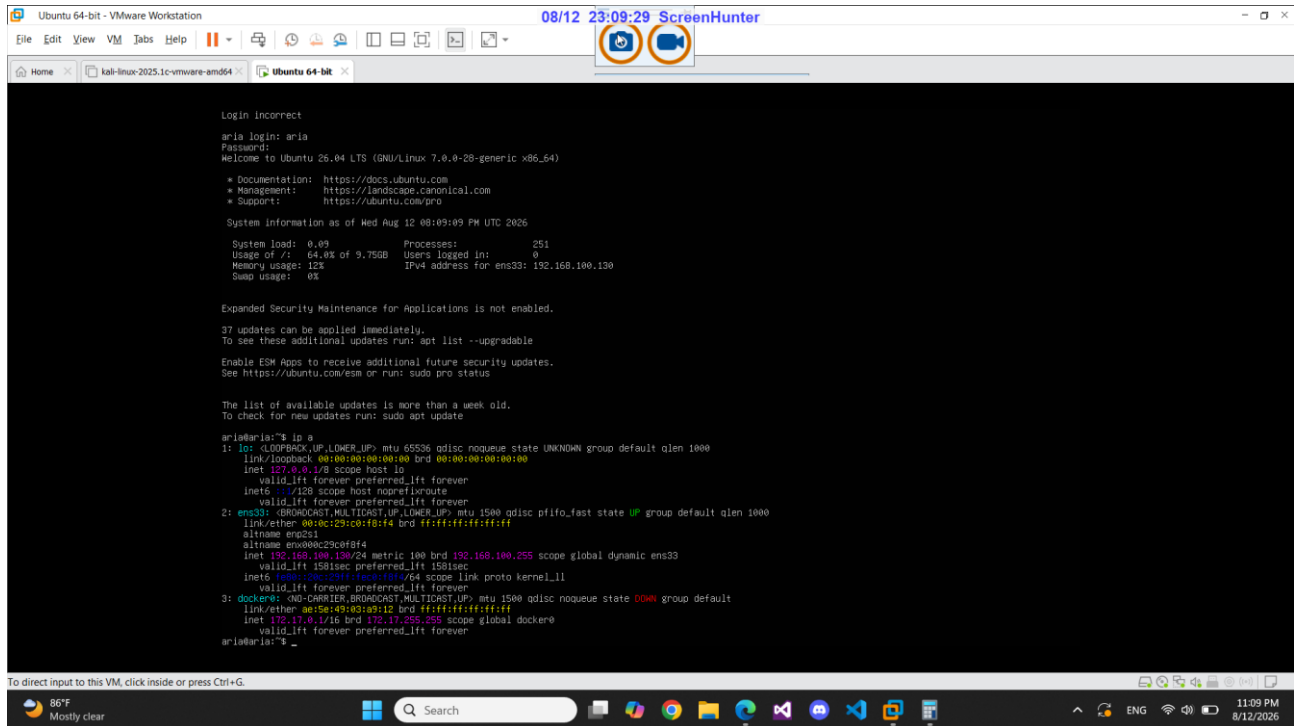
The network should remain isolated from untrusted networks during testing because DVWA and Metasploitable are intentionally vulnerable systems.

5. Step 2 — Verify the Ubuntu/DVWA Target IP

On the Ubuntu virtual machine hosting DVWA, the network interface was inspected with the `ip a` command. The active interface `ens33` is configured with 192.168.100.130/24.

Observed configuration:

- Interface: `ens33`
- IPv4 address: 192.168.100.130/24
- Broadcast: 192.168.100.255
- Interface state: UP



```

Login incorrect
aria login: aria
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-28-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Wed Aug 12 08:09:09 PM UTC 2026

System load:  0.09          Processes:      251
Usage of /:   64.6% of 9.75GB Users logged in:   0
Memory usage: 12%          IPv4 address for ens33: 192.168.100.130
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

37 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

aria@aria:~$ ifconfig
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noforwardroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:c0:f0:f4 brd ff:ff:ff:ff:ff:ff
    altname ens33
    altname enx00c29c0f0f4
    inet 192.168.100.130/24 metric 100 brd 192.168.100.255 scope global dynamic ens33
        valid_lft 1581sec preferred_lft 1581sec
    inet6 fe80::c0c1:2011:1ec3:1844/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
3: dockere: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether ae:5e:49:a3:a3:12 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global dockere
        valid_lft forever preferred_lft forever
aria@aria:~$

```

Figure 2. Ubuntu/DVWA host — verified IPv4 configuration.

6. Step 3 — Verify the Metasploitable 2 IP

On the Metasploitable 2 target, the network configuration was checked with ifconfig. The eth0 interface is up and has the IPv4 address 192.168.100.132 with a /24 subnet mask.

Observed configuration:

- Interface: eth0
- IPv4 address: 192.168.100.132
- Subnet mask: 255.255.255.0 (/24)
- Broadcast: 192.168.100.255
- Interface state: UP and RUNNING

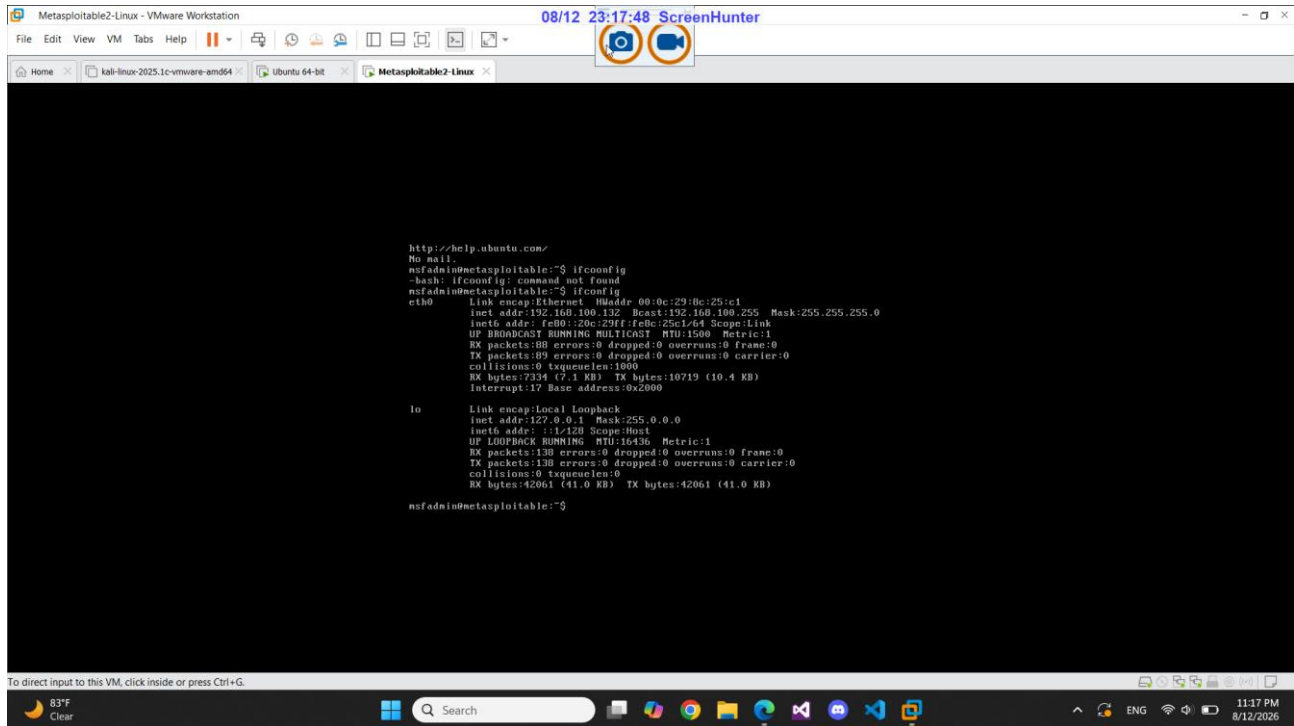


Figure 3. Metasploit2 — verified eth0 network configuration.

7. Step 4 — Verify DVWA Availability

From Kali Linux, the DVWA web application was opened in Firefox using the target address 192.168.100.130 on port 8081. The DVWA welcome page loaded successfully, confirming that the web application is reachable over the lab network.

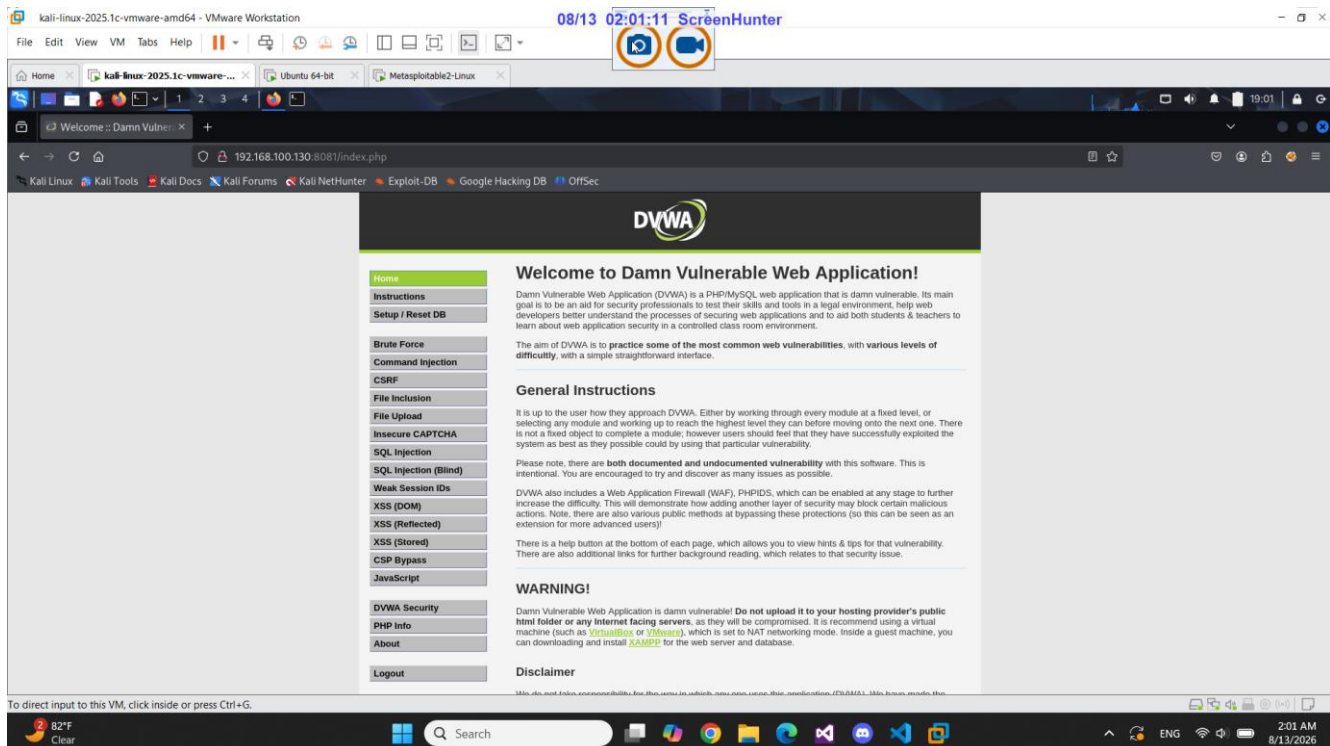


Figure 4. DVWA web interface successfully loaded from Kali Linux.

8. Step 5 — Test Network Connectivity from Kali

Connectivity was tested from the Kali Linux VM using ICMP ping. Both target addresses responded successfully with 0% packet loss.

Target	Command	Result	Status
Ubuntu/DVWA	ping -b 192.168.100.130	11 packets transmitted, 11 received, 0% loss	PASS
Metasploitable 2	ping -b 192.168.100.132	6 packets transmitted, 6 received, 0% loss	PASS

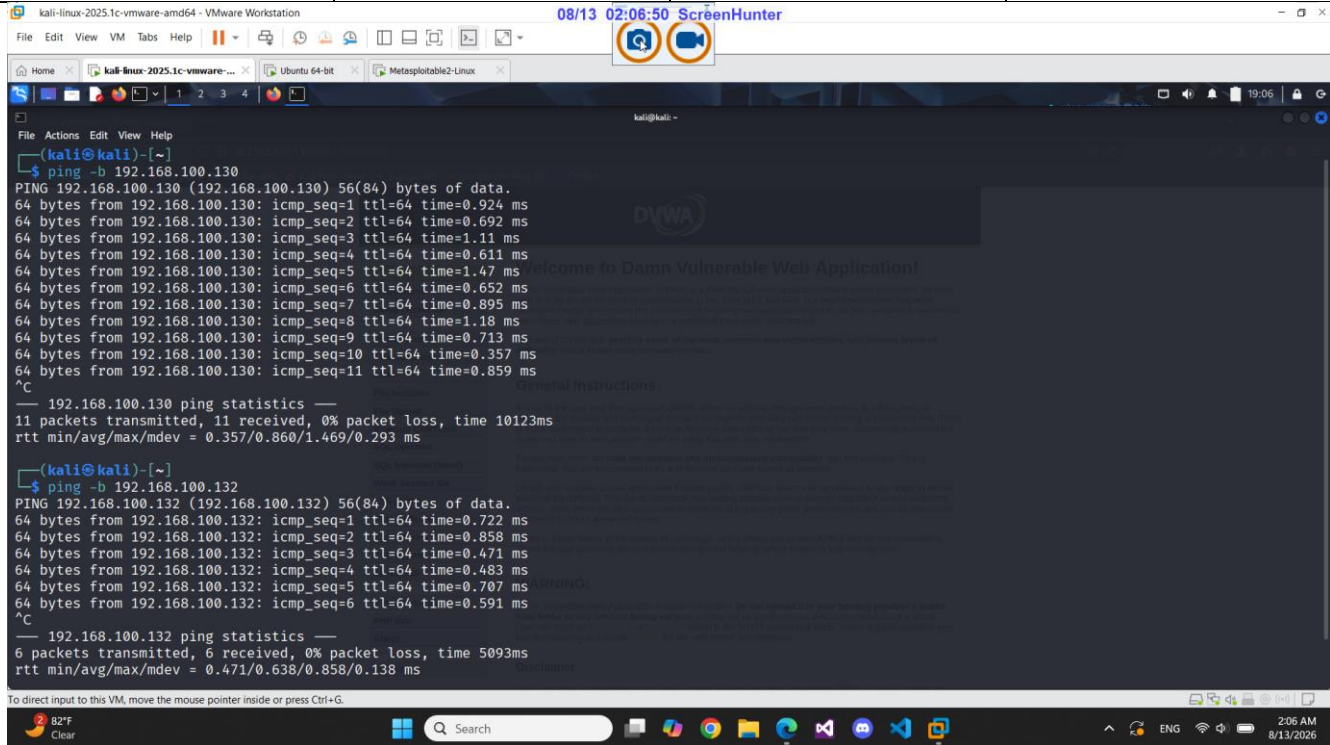


Figure 5. Kali Linux — successful ICMP connectivity tests to both targets.

9. Final Verification Checklist

Check	Evidence / Result
Kali Linux is connected to the lab network	Verified by topology and connectivity tests
Ubuntu/DVWA IP is configured	192.168.100.130/24
Metasploitable 2 IP is configured	192.168.100.132/24
DVWA is accessible	Web interface loaded successfully
Kali → DVWA connectivity	PASS — 0% packet loss
Kali → Metasploitable connectivity	PASS — 0% packet loss
Lab network is isolated	Private 192.168.100.0/24 VAPT environment

10. Safety and Scope

All vulnerability-testing activities should be limited to the systems and networks explicitly authorized for the lab. Because DVWA and Metasploitable 2 are intentionally insecure, the VMs should not be exposed directly to the

public Internet. Snapshots are recommended before major changes so the lab can be restored to a known-good state.

11. Conclusion

The VAPT lab was successfully established as an isolated virtual environment. Kali Linux is able to communicate with both target systems, the Ubuntu host exposes DVWA at 192.168.100.130:8081, and Metasploitable 2 is reachable at 192.168.100.132. The environment is therefore ready for authorized penetration-testing and vulnerability-assessment exercises.